Homework Practice Sheet (Intro ML / Neural Nets)
Forward Pass, Backpropagation, Gradient Descent + Concepts

Instructions

Show your work. Unless otherwise stated, use:

$$\sigma(z) = \frac{1}{1+e^{-z}}, \qquad \mathrm{ReLU}(z) = \max(0, z),$$

$$L_{\mathrm{MSE}} = \frac{1}{2}(\hat{y} - y)^2, \qquad \text{and} \qquad w \leftarrow w - \eta \frac{\partial L}{\partial w}.$$

For conceptual questions, answer in 1–4 sentences.

Part A: Numeric Problems (Forward Pass / Backprop / Updates)

1. **Forward pass (2 inputs, linear neuron).**
   Given $x = \begin{bmatrix} 2 \\ -1 \end{bmatrix}$, $w = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$, $b = -2$. Compute:

$$z = w^\top x + b.$$

2. **Forward pass (sigmoid neuron).**
   Given $x = 4$, $w = -0.5$, $b = 1$. Compute:

$$z = wx + b, \qquad \hat{y} = \sigma(z).$$

3. **Forward pass (tiny 1-hidden-layer net).**
   Let

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad W_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix}, \quad b_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

   Compute:

$$z_1 = W_1 x + b_1, \qquad h = \mathrm{ReLU}(z_1).$$

   Then with

$$W_2 = \begin{bmatrix} 2 & -1 \end{bmatrix}, \quad b_2 = 0,$$

   compute:

$$y = W_2 h + b_2.$$

4. **MSE loss.**
   Given prediction $\hat{y} = -1$ and target $y = 2$, compute:
$$L = \frac{1}{2}(\hat{y} - y)^2.$$

5. **Backprop warm-up (MSE derivative).**
   For $L = \frac{1}{2}(\hat{y} - y)^2$, compute $\frac{\partial L}{\partial \hat{y}}$ in terms of $\hat{y}$ and y.

6. **Backprop for a linear neuron (2 inputs).**
   Model: $\hat{y} = w^\top x + b$, loss $L = \frac{1}{2}(\hat{y} - y)^2$.
   Given:
$$x = \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad w = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad b = 1, \quad y = 0.$$
   Compute $\frac{\partial L}{\partial w_1}$, $\frac{\partial L}{\partial w_2}$, and $\frac{\partial L}{\partial b}$.

7. **One gradient descent update (linear neuron).**
   Use the same setup as Problem 6 with learning rate $\eta = 0.2$. Compute the updated w and b.

8. **ReLU backprop (single neuron).**
   Let $z = wx + b$, $a = \mathrm{ReLU}(z)$, and $L = \frac{1}{2}(a - y)^2$.
   Given $x = -1$, $w = 2$, $b = -1$, $y = 3$.
   Compute z, a, and then $\frac{\partial L}{\partial w}$ and $\frac{\partial L}{\partial b}$. (Be careful about the ReLU derivative.)

9. **Two-step gradient descent (linear regression).**
   Data point $(x, y) = (2, 1)$, initialize $w_0 = 0$, $b_0 = 0$, learning rate $\eta = 0.1$.
   Run **two** steps of gradient descent using:
$$\hat{y} = wx + b, \quad L = \frac{1}{2}(\hat{y} - y)^2.$$
   Report (w_1, b_1) and (w_2, b_2).

10. **Softmax probabilities (3 classes).**
    Given logits $s = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$. Compute $\mathrm{softmax}(s)$. (You may leave answers in terms of e.)

Part B: Conceptual Questions (Short Answers)

11. **Forward pass vs. backprop.**
    In your own words, what happens during the forward pass, and what happens during backpropagation?

12. **Why do we need activation functions?**
    What problem would occur if a neural network used *no* nonlinear activation functions (only linear layers)?

13. **Learning rate intuition.**
    What can happen if the learning rate is (a) too large, or (b) too small?

14. **Vanishing gradients (high level).**
    What does "vanishing gradients" mean, and why can it make training deep networks difficult?

15. **ReLU vs. sigmoid (practical).**
    Give one reason ReLU is often preferred over sigmoid for hidden layers.

1. $z = W^T x + b$

$= [1\ 3] \begin{bmatrix} 2 \\ -1 \end{bmatrix} + (-2)$

$= 2 - 3 - 2$

$= -3$

2. $z = 4 \times -0.5 + 1$

$= -2 + 1$

$= -1$

$y = \sigma(-1)$

$= \frac{1}{1+e^{(-1)}} = \frac{1}{1+e}$

3. $z_1 = \begin{bmatrix} 1 & -1 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} 1-2 \\ 2+0 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix}$

$= \begin{bmatrix} -1 \\ 3 \end{bmatrix}$

$h = \mathrm{ReLU}(z_1) = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$

$y = W_2 h + b_2$

$= [2\ -1] \begin{bmatrix} 0 \\ 3 \end{bmatrix} + 0$

$= -3$

4. $\hat{y} = -1, \ y = 2$

$L = \frac{1}{2}(-1-2)^2$

$= \frac{1}{2}(-3)^2$

$= \frac{9}{2}$

5. $L = \frac{1}{2}(\hat{y} - y)^2$

$\frac{\partial L}{\partial \hat{y}} = 2 \cdot \frac{1}{2}(\hat{y} - y) \cdot$

$= \hat{y} - y$

6. $L = \frac{1}{2}(\hat{y} - y)^2$

$\hat{y} = [0\ 1]\begin{bmatrix}1\\-2\end{bmatrix} + 1$

$= -2+1$

$= -1$

$\frac{\partial L}{\partial y} = 2 \cdot \frac{1}{2}(\hat{y} - y)$

$= \hat{y} - 1$

$= -1 - 0$

$= -1$

$\frac{\partial L}{\partial w_1} = (\hat{y} - y) \cdot x_1 = -1 \cdot 1 = -1$

$\frac{\partial L}{\partial w_2} = (\hat{y} - y) \cdot x_2 = (-1) \cdot -2 = 2$

$\frac{\partial L}{\partial b} = \hat{y} - y = -1$

7. $w_1' = w_1 - \eta \frac{\partial L}{\partial w_1}$

$= 0 - 0.2(-1)$

$= 0.2$

$w_2' = w_2 - \eta \frac{\partial L}{\partial w_2} = 1 - 0.2 \cdot 2$

$= 0.6$

$b' = b - \eta \frac{\partial L}{\partial b} = 1 - 0.2(-1)$

$= 1.2$

$W' = \begin{bmatrix}0.2\\0.6\end{bmatrix}, \quad b' = 1.2$

8. $z = 2 \cdot -1 - 1$

$= -3$

$a = \mathrm{BeLU}(-3)$

$= 0$

$L = \frac{1}{2}(a - y)^2$

$\frac{\partial L}{\partial w} = \frac{1}{2} \cdot 2(a - y) \cdot 0 \cdot x$

$= 0$

$\frac{\partial L}{\partial b} = \frac{1}{2} \cdot 2(a - y) \cdot 0 \cdot 1$

$= 0$

9. $\hat{y} = 0(2) + 0$
$= 0$

$L = \frac{1}{2}(\hat{y} - y)^2$

$\frac{\partial L}{\partial w} = (\hat{y} - y)(x)$
$= (0-1)(2)$
$= -2$

$\frac{\partial L}{\partial b} = (\hat{y} - y)(1)$
$= -1$

$w_1 = 0 - 0.1(-2)$
$= 0.2$

$b_1 = 0 - 0.1(-1)$
$= 0.1$

$\hat{y} = 0.2(2) + 0.1$
$= 0.5$

$\frac{\partial L}{\partial w} = (\hat{y} - y)(x)$
$= (0.5 - 1)(2)$
$= -1$

$\frac{\partial L}{\partial b} = (\hat{y} - y)(1)$
$= -0.5$

$w_2 = 0.2 - 0.1(-1)$
$= 0.3$

$b_2 = 0.1 - 0.1(-0.5)$
$= 0.15$

10. $p = \begin{pmatrix} \frac{1}{1+e+e^2} \\ \frac{e}{1+e+e^2} \\ \frac{e^2}{1+e+e^2} \end{pmatrix}$

11. During forward pass input date flows through the network and each layer, applies its weights biases, and activation functions. This produces final prediction and loss value.

During backpropagation, we want to find the gradient of the loss with respect to every parmeter. We use chain rule in reverse

12. We need them to introduce non-linearity into the output of a neuron. Otherwise, every layer will perform a linear activation. We won't be able to learn any nonlinear decision boundary.

13. We will get unstable training, and oscillating weight updates. We will also overshoot the optimal minimum.

If it is too small, training converges very slowly, and we can get stuck with inadequate local minima.

14. Sometimes, gradients shrink when they are backpropagated. If this happens a lot then early layers will receive near-zero gradients.

15. ReLU gradient is 0 or 1, so it is less susceptible to shrinking gradients.